

Avinash Tiwari, Ph.D. Student

Completion Expected: May, 2026

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Current Affiliations

- 2021 – 📌 Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2022 – 📌 LIGO Scientific Collaboration and LIGO-India Scientific Collaboration (LISC).

Education

- 2021 – 📌 **Ph.D.**, Inter-University Centre for Astronomy and Astrophysics, Pune.
- 2019 – 2021 📌 **M.Sc. Physics**, Indian Institute of Technology Guwahati, Guwahati.
- 2016 – 2019 📌 **B.Sc. Physics, Mathematics, and Computer Science**, University of Allahabad, Prayagraj.

Research Interests

- 📌 Bayesian inference and GW data analysis, especially, but not restricted to, Fisher Matrix Analysis, Parameter Estimation, and Hierarchical (Hyperparameter) Inference.
- 📌 GW waveform modeling, especially line-of-sight acceleration (LOSA) waveforms
- 📌 Probing the compact binary coalescence (CBC) environments using GWs.
- 📌 GW population studies.
- 📌 Cosmology.

Skills

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| Languages | 📌 English, Hindi. |
| Coding | 📌 Python, C, $\text{\LaTeX}$, Mathematica. |
| Web Dev | 📌 GITHUB PAGES. |
| Packages | 📌 BILBY, BILBY_PIPE, BILBY_TGR, GWPOPULATION, EMCEE, DYNESTY. |
| Software Development | 📌 BILBY_TGR — Developed the LOSA pipeline within the package. |

Accomplishments

- 📌 **Graduate Aptitude Test in Engineering** (GATE 2021): All India Rank 209 in Physics.
- 📌 **Junior Research Fellowship** (JRF)-CSIR: All India Rank 89 in Physical Sciences, Joint CSIR - UGC NET JUNE 2020 (held in November 2020).
- 📌 **IASC** (November 9 – December 4, 2020): Member of the team “Delhi University” in the “DST-Rajasthan Asteroid Search Campaign” organised by DST-Rajasthan in collaboration with the International Astronomical Search Collaboration (IASC).
- 📌 **IASC** (August 12 – September 8, 2020): Member of the team “IIT Guwahati” in the “DST-Rajasthan Asteroid Search Campaign” organised by DST-Rajasthan in collaboration with the International Astronomical Search Collaboration (IASC).

Accomplishments (continued)

- **Institute Merit-cum-Means (McM) Scholarship:** awarded in both years of M.Sc. (2019-21) by the India Institute of Technology Guwahati on merit-cum-means basis for being in the top 25%.
- **Joint Admission Test for M.Sc. (JAM) 2019:** All India Rank 236 in Physics.

Conferences, Workshops, and Colloquia

- **GW - Detector Characterization Workshop** (December 15 - 19, 2025), IUCAA, Pune.
- **GR24 - Amaldi16** (July 14 - 18, 2025): Talk on "Profiling stellar environments of gravitational wave sources".
- **AEI Colloquium** (August 22, 2024), AEI Hanover: "Profiling stellar environments of gravitational wave sources".
- **New ideas on the ORIGIN of BLACK HOLE MERGERS** (August 12 – August 16, 2024), NBI Copenhagen: Invited Talk on "Profiling stellar environments of gravitational wave sources".
- **ICGC-2023** (December 6 – 9, 2023): Talk on "Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers".
- **Amaldi15** (July 17 - 21, 2023): Talk on "Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers".
- **Discussion Meeting on LUNAR GRAVITATIONAL-WAVE DETECTION** (April 17 - 20, 2023), Ramanujan Lecture Hall, ICTS Bengaluru.
- **IAGRG32** (December 19 - 21, 2022): Poster presentation and a Flash Talk on "Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves".

Other Talks / Seminars

- **Last Friday Talks at IUCAA** (August 29, 2025): "Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries".
- **Last Friday Talks at IUCAA** (September 27, 2024): "Profiling stellar environments of gravitational wave sources".
- **Last Friday Talks at IUCAA** (July 28, 2023): "Probing globular clusters and other astrophysical environments with gravitational waves emitted by accelerated compact binary mergers".
- **Last Friday Talks at IUCAA** (March 31, 2023): "Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves".
- **Online Seminar at ICTS, Bengaluru** (December 15, 2022): "Waltzing binaries: probing line-of-sight acceleration of merging compact objects with gravitational waves".

Courses Tutored

- PHY-IC481 - General Relativity at IUCAA.
- PHY-IE485 - Elective course - Cosmology at IUCAA.

Mentoring

- Sudhir Gholap - on Eccentricity Evolution Consistency Test (EECT).
- Md Asim Ansari - extending the LOSA waveforms to eccentric binaries.